

SVICHTOV, Bulgaria

WMO: 155330

Lat: 43.617N

Lon: 25.350E

Elev: 85

StdP: 14.65

Time Zone: 2.00 (EUE)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.7	16.5	9.0	8.8	12.7	13.5	11.0	17.1	37.6	37.0	33.9	37.9	4.0	50	0.448

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	94.2	74.6	91.0	73.5	88.2	72.0	76.5	91.1	74.7	88.4	73.2	86.1	4.6	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.7	117.5	86.0	70.0	110.9	83.9	68.5	105.1	81.8	40.0	90.8	38.3	88.5	36.9	86.3	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
33.0	26.9	21.8	7.4	100.1	6.4	4.5	2.8	103.3	-1.0	106.0	-4.6	108.5	-9.3	111.8
			6.8	80.2	6.4	2.1	2.2	81.7	-1.5	82.9	-5.1	84.1	-9.7	85.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.6	32.0	36.6	46.3	56.2	66.1	73.2	77.3	76.8	67.2	55.6	44.3	33.8	(6)
	DBStd	17.61	9.08	9.75	8.54	6.49	6.00	5.70	5.17	5.16	6.09	7.34	7.46	8.69	(7)
	HDD50	1873	560	380	175	21	0	0	0	0	0	31	202	504	(8)
	HDD65	4673	1023	796	581	269	61	7	0	1	47	300	621	967	(9)
	CDD50	3902	1	5	60	208	500	696	846	830	516	206	32	2	(10)
	CDD65	1224	0	0	1	6	95	253	381	366	113	9	0	0	(11)
	CDH74	10648	0	0	4	68	693	2153	3628	3329	726	46	1	0	(12)
	CDH80	4106	0	0	0	14	158	786	1560	1399	186	3	0	0	(13)
(14) Wind	WSAvg	5.8	5.7	6.7	7.5	6.4	6.1	5.8	5.8	4.7	5.2	4.7	5.1	5.5	(14)
(15) Precipitation	PrecAvg	22.2	1.7	1.3	1.5	1.7	2.4	2.4	2.0	1.6	2.2	1.9	1.4	1.9	(15)
	PrecMax	35.7	4.7	3.9	3.1	3.3	5.4	6.7	6.6	5.1	6.5	4.9	3.9	6.2	(16)
	PrecMin	13.1	0.1	0.1	0.0	0.1	0.6	0.2	0.3	0.2	0.2	0.0	0.0	0.2	(17)
	PrecStd	5.4	1.0	0.8	0.8	0.8	1.1	1.5	1.6	1.3	1.6	1.3	0.9	1.3	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.5	62.7	72.9	81.7	87.9	94.7	99.7	98.5	89.5	79.2	68.2	59.0	(19)
		MCWB	48.0	51.2	56.8	64.8	69.1	74.8	76.6	75.6	70.5	67.5	58.8	52.0	(20)
	2%	DB	51.7	57.5	68.2	74.5	84.1	90.7	94.7	93.9	84.8	74.2	62.6	53.8	(21)
		MCWB	46.0	48.8	54.4	60.5	67.6	73.8	74.9	74.5	69.1	64.2	56.1	48.6	(22)
	5%	DB	47.6	53.8	64.1	70.5	81.1	87.7	91.3	90.5	81.4	69.8	58.5	48.2	(23)
		MCWB	43.3	47.0	52.3	58.2	66.5	72.0	73.6	73.3	67.5	61.6	54.0	45.1	(24)
	10%	DB	43.9	49.8	60.0	67.3	77.8	84.8	88.1	87.6	78.0	66.3	54.8	44.4	(25)
		MCWB	40.6	44.3	50.2	56.3	65.1	70.4	72.2	71.8	65.8	59.7	51.0	41.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.1	52.8	58.8	65.9	71.8	77.3	78.4	79.4	72.4	68.0	59.9	52.5	(27)
		MCDB	54.9	60.3	69.6	78.2	83.5	91.3	94.9	94.6	86.6	77.6	66.3	58.5	(28)
	2%	WB	46.4	49.6	55.6	61.7	69.5	75.0	76.4	76.5	70.1	65.1	56.8	48.7	(29)
		MCDB	50.8	56.5	65.8	72.3	81.3	88.3	92.1	90.4	82.6	72.6	61.5	53.2	(30)
	5%	WB	43.5	47.1	53.2	59.4	67.5	73.2	74.7	74.6	68.3	62.5	54.4	45.5	(31)
		MCDB	47.5	53.0	62.9	68.7	78.8	85.7	88.9	87.8	79.5	68.6	58.0	48.1	(32)
	10%	WB	40.7	44.6	50.6	57.4	65.7	71.4	73.1	72.9	66.7	60.1	51.8	42.2	(33)
		MCDB	43.7	49.6	59.3	65.4	75.9	82.9	86.1	85.4	76.6	65.5	54.6	44.0	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.7	11.6	15.6	17.2	18.7	18.6	18.8	19.1	16.8	13.9	9.7	8.4	(35)
		MCDBR	13.8	19.0	22.9	22.5	23.5	22.6	23.4	23.2	22.4	19.1	15.2	15.2	(36)
	5% WB	MCWBR	9.9	12.5	12.6	11.6	10.8	9.7	9.5	9.5	10.3	11.0	10.3	11.5	(37)
		MCWBR	13.4	18.2	22.0	20.8	21.8	21.9	22.3	22.1	20.8	17.9	13.7	14.3	(38)
(40) Clear-Sky Solar Irradiance	taub		0.348	0.380	0.419	0.471	0.446	0.454	0.464	0.477	0.443	0.405	0.374	0.352	(40)
	taud		2.281	2.210	2.150	2.019	2.167	2.183	2.162	2.122	2.183	2.270	2.313	2.292	(41)
	Ebn at Noon		239	252	257	252	263	261	257	248	246	240	228	225	(42)
	Edh at Noon		28	35	43	52	46	46	46	47	41	33	27	25	(43)
(44) All-Sky Solar Radiation	RadAvg		469	725	1086	1506	1864	2043	2068	1851	1347	874	499	381	(44)
	RadStd		70	93	104	154	150	139	137	112	104	98	70	53	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	+2.90	+2.48	+2.00	N/A	N/A	N/A	N/A	
(47)	Regional (48 neighbors)	+0.81	N/A	N/A	N/A	+0.76	N/A	N/A	-252	+155	N/A	

Nomenclature: See separate page